

Marginal Matching Does Not License Factorized Sampling: Auditing Conditional Style Leakage in Factorized Generative Models

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Abstract

Factorized generative models regularize a style latent z_s toward a fixed prior with a marginal statistic, $q(z_s) \approx \mathcal{N}(0, I)$, and treat the result as a certificate that z_s carries no class information. The certificate does not hold. Matching the marginal constrains nothing about the class conditionals $q(z_s | y)$, so z_s can be exactly Gaussian in aggregate while remaining maximally informative about the label. We show this gap is one of four terms in an exact decomposition of what a factorized sampler must match, and that eliminating it is necessary but not sufficient for the sampling procedure the factorization exists to support. Empirically, a case-study model and four reference latent baselines permit a linear probe to recover the label from z_s at 74–100% (chance 10%); our case-study model reaches 99.15% clustering accuracy while externally evaluated class-conditional generation succeeds 16% of the time. The leakage survives six one-at-a-time perturbations of capacity, curriculum, prior geometry, and supervision on two datasets. Four remedies attacking the leakage through different mechanisms span 21–46% probe recovery while leaving the within-class dependence proxy essentially unchanged. A post-hoc conditional prior raises external generated-class accuracy to 0.97 on MNIST without retraining but reaches only 0.41 on CIFAR-10; an empirical style bank reaches 0.88 on CIFAR-10. The methodological point: no divergence computed on $q(z_s)$ alone can certify $z_s \perp y$, and reporting marginal statistics does not verify the property practitioners claim from them.

Introduction

Matching an aggregate posterior to a Gaussian is a statement about a marginal. Class-invariant style is a statement about a family of conditionals. The two are different claims, and the first does not imply the second: $q(z_s)$ can be exactly $\mathcal{N}(0, I)$ while every $q(z_s | y=k)$ occupies its own region of latent space, so long as the regions average out. A regularizer that only ever sees $q(z_s)$ cannot detect this, and therefore cannot prevent it.

The gap has teeth because of how factorized models generate. During training the decoder only ever receives (z_c, z_s) drawn from the same image. If $q(z_s | y=k)$ differs across classes, and nothing in the objective stops it, the decoder

is free to learn class-specific interactions between the two codes. At generation time a class prior supplies z_c for class k while z_s comes from the marginal, which averages over all classes. The resulting pair is one the decoder never saw. We call the failure *conditional style leakage*.

What makes it dangerous is that the metrics a practitioner would check are all blind to it. Reconstruction is measured on the posterior manifold, where z_s comes from a real image and the learned interaction is exactly the one being exercised, so it looks fine. Global MMD is a marginal statistic by construction. Clustering accuracy reads only z_c . A model can therefore pass every standard diagnostic and still fail completely at the one capability the factorization was built to provide.

Our starting point is that this is not a defect of any particular loss but a structural property of factorized sampling. We give an exact decomposition of the mismatch such a sampler incurs (Theorem 1) into four non-negative terms: global style-prior mismatch, conditional style leakage, semantic-prior mismatch, and within-class dependence between the two codes. Valid sampling requires all four to vanish, so removing the leakage is necessary but not sufficient, and the latent mismatch upper-bounds the image-space mismatch but not conversely. That asymmetry is what makes a decoder-level intervention necessary rather than redundant.

We then audit a case-study model and four reference latent baselines with finite-sample proxies for the four terms, and use F-CS-WAE (Factorized Class-Structured Spherical Cauchy WAE) as a concentrated case study. We use it to exhibit and repair the failure, not to argue it beats the baselines it is compared against. The four reference families share one diagnostic architecture and budget, while F-CS-WAE is evaluated in its native configuration, so the cross-model table supports within-model auditing rather than a ranking. F-CS-WAE is label-guided by design; the proposed per-class style MMD reuses the labels already required by its semantic objective and therefore adds no annotation requirement.

Contributions. (1) An auditing framework for factorized generative models that separates marginal prior fit, conditional leakage, semantic-prior mismatch, and within-class dependence, supported by a structural KL decomposition. (2) A conditional audit protocol built from finite-sample proxies for those terms, applied to a case-study model and reference

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latent baselines. (3) A decoder-level validation study showing when information detected by the audit is actually used during recombination, together with a six-way robustness check on two datasets, showing no single design choice accounts for the leakage while its magnitude is strongly capacity-dependent on CIFAR-10. (4) An evaluation of representation- and sampling-level repairs, including a negative transfer result.

Related Work

We organize prior work by a single question: does the method certify conditional invariance $z_s \perp y$, or only a statistic compatible with $z_s \not\perp y$?

VAEs (Kingma and Welling 2013) impose a per-example KL toward a fixed prior; WAEs (Tolstikhin et al. 2017) relax this to a divergence between the aggregated posterior and the prior, computed with kernel two-sample tests (Gretton et al. 2012) or projected distances (Kolouri et al. 2018). The VAE penalty is not simply an aggregate regularizer in disguise: averaged over the data it splits exactly into $\mathbb{E}_{q(x)}[\text{KL}(q(z|x)||p(z))] = I_q(X; Z) + \text{KL}(q(z)||p(z))$ (Hoffman and Johnson 2016), an information-capacity term plus an aggregate-mismatch term of the kind WAE-MMD targets directly. Neither certifies $I(z_s; y) = 0$. The capacity term bounds what Z encodes about X in general, not about the label, and the aggregate term is precisely the marginal statistic Proposition 1 shows is insufficient.

β -VAE (Higgins et al. 2017), β -TCVAE (Chen et al. 2018), FactorVAE (Kim and Mnih 2018), and DIP-VAE (Kumar, Sattigeri, and Balakrishnan 2018) penalize total correlation or aggregate moments. These constrain dependence among latent *dimensions*, not between a designated style code and the label. More generally, unsupervised disentanglement is not identifiable without inductive biases in the model or data (Locatello et al. 2019); weakly supervised pairs can supply such a bias (Locatello et al. 2020), but a bias toward factorization is not itself a certificate of the particular invariance $z_s \perp y$. Semi-supervised VAEs (Kingma et al. 2014) and CVAE (Sohn, Lee, and Yan 2015) condition generation on the label, sidestepping the need for z_c to encode identity, but do not by themselves test whether a residual latent remains label-informative.

Content–style separation predates deep generative models (Tenenbaum and Freeman 2000). Modern split-latent models use class labels, grouped observations, pairwise similarity, mutual-information penalties, or latent optimization to divide specified from residual variation (Mathieu et al. 2016; Bouchacourt, Tomioka, and Nowozin 2018; Jha et al. 2018; Klys, Snell, and Zemel 2018; Zheng and Sun 2019; Ilse et al. 2020; Gabbay and Hoshen 2020). These works establish that explicit inductive biases can improve recombination; they also make a plain VAE latent an inappropriate surrogate for a declared class-invariant style code. Most directly, Ridgeway and Mozer (2018) introduce leakage filtering, probe class from a style posterior, and evaluate content–style recombination on MNIST and more complex domains. Our novelty is therefore neither the first observation that content can leak into style nor the algebraic chain rule in isolation. It is the

auditing framework that uses the decomposition to separate four distinct failure sources, connects finite-sample latent diagnostics to the claim practitioners make from marginal matching, and validates at decoder level whether detected leakage changes recombination and sampling.

Image-translation systems operationalize the same recombination requirement: MUNIT (Huang et al. 2018) and DRIT (Lee et al. 2018) combine content with sampled style, while DMIT (Yu et al. 2019) exposes the generator to random cross-domain combinations during training. Their notion of domain-specific style is task-dependent—the domain may itself be the conditioning attribute—but their training strategies highlight the same train–sample support issue. Recent methods add other inductive biases: V3 exploits variance–invariance patterns across domains (Wu et al. 2025), while SCFlow learns invertible merging from combinatorial style–content coverage (Ma et al. 2025). These methods address how to learn a separation. Reference-guided diffusion work also calls unwanted reference content carried by style features “content leakage” and suppresses it by masking (Zhu et al. 2025); our leakage is instead label information in a learned style latent. We ask when such a latent can be sampled independently.

Several invariance methods directly target $z_s \not\perp y$. The variational fair autoencoder (Louizos et al. 2015) matches nuisance-conditional posteriors to a common target; the per-class style MMD in Section is a direct instantiation of that idea, not a new mechanism. Gradient reversal (Ganin and Lempitsky 2015), Fader Networks (Lample et al. 2017), and information-theoretic invariance (Moyer et al. 2018) remove specified information through different objectives. They primarily optimize representation invariance; our decomposition asks the additional question of whether removing $z_s \not\perp y$ suffices for the independent sampler. It does not when the other three terms remain.

Deep generative clustering (DEC (Xie, Girshick, and Farhadi 2015), IDEC (Guo et al. 2017), VaDE (Jiang et al. 2016)) evaluates via ACC/NMI/ARI on the cluster variable, a protocol that cannot see this failure because it never inspects z_s . Our case study makes the point concrete: 99.15% clustering accuracy on the very checkpoint whose class-conditional generation reaches 16% external Gen-ACC. For the semantic variable we use a Spherical Cauchy prior with a Möbius reparameterization (Sablica and Hornik 2025), a heavier-tailed alternative to the von Mises-Fisher posterior of S-VAE (Davidson et al. 2018); this is an implementation choice, not a claim about the leakage, which is defined independently of how z_c is parameterized.

Factorized Sampling and Its Mismatch

Let $\mathcal{D} = \{(x_i, y_i)\}$ have K classes. A factorized model encodes each input into a semantic variable z_c and a style variable z_s , and regularizes the latter by

$$\mathcal{L}_{\text{style}} = D(q(z_s), \mathcal{N}(0, I)), \quad (1)$$

for some aggregate divergence D , with $q(z_s) = \mathbb{E}_y[q(z_s | y)]$. Write $p(z_s) := \mathcal{N}(0, I)$ for the style prior. Together with a class-conditional semantic prior $p(z_c | y=k)$ these

define the *factorized sampler*: draw $z_c \sim p(z_c \mid y=k)$ and $z_s \sim p(z_s)$ independently, which is the naive class-conditional sampling procedure in standard use.

Marginal matching certifies nothing

Proposition 1 (No certificate). *Let $z_s \sim \mathcal{N}(0, I)$ on $\mathbb{R}^{d_s}$ and let $\{A_k\}_{k=1}^K$ be any measurable partition with $P(z_s \in A_k) = 1/K$. Define $y = k \iff z_s \in A_k$. Then $q(z_s) = \mathcal{N}(0, I)$ exactly, so $D(q(z_s), \mathcal{N}(0, I)) = 0$ for every divergence D ; and $I(z_s; y) = H(y) = \log K$, the maximum possible.*

The proof is immediate: y is defined post hoc as a measurable function of z_s , so the law of z_s is untouched, while $H(y \mid z_s) = 0$ makes the mutual information maximal. The construction is deliberately adversarial and we do not claim trained decoders produce anything like it. The point is narrow and worst-case: no divergence computed on $q(z_s)$ alone, MMD or otherwise, can rule out z_s being maximally class-informative, however the dependence happens to be shaped.

What valid factorized sampling requires

Proposition 1 is the extremal case of term (2) below. The following decomposition covers any encoder, not just that construction.

Theorem 1 (Factorized-sampling mismatch). *Let y be uniform on $\{1, \dots, K\}$ and define $M_{\text{fact}} := \mathbb{E}_y[\text{KL}(q(z_c, z_s \mid y) \parallel p(z_c \mid y)p(z_s))]$. Then*

$$M_{\text{fact}} = \underbrace{I_q(z_c; z_s \mid y)}_{(4) \text{ within-class dep.}} + \underbrace{\mathbb{E}_y[\text{KL}(q(z_c \mid y) \parallel p(z_c \mid y))]}_{(3) \text{ semantic-prior mismatch}} + \underbrace{I_q(z_s; y)}_{(2) \text{ style leakage}} + \underbrace{\text{KL}(q(z_s) \parallel p(z_s))}_{(1) \text{ style-prior mismatch}},$$

and all four terms are non-negative.

The proof is a chain-rule expansion applied twice, given in the supplementary material. Two consequences matter.

Corollary 1 (Sampling validity). *$q(z_c, z_s \mid y=k) = p(z_c \mid y=k)p(z_s)$ for a.e. k if and only if $M_{\text{fact}} = 0$, i.e. iff all four terms vanish. Class-invariant style (term 2) is thus necessary but not sufficient; it must be accompanied by a matched style prior, a matched semantic prior, and within-class independence of z_c and z_s .*

Corollary 2 (Decoder pushforward). *For a decoder Dec with pushforward $\text{Dec}_\#$, $\mathbb{E}_y[\text{KL}(\text{Dec}_\#q(\cdot \mid y) \parallel \text{Dec}_\#p(z_c \mid y)p(z_s))] \leq M_{\text{fact}}$ by the data-processing inequality.*

Corollary 2 runs one way. A small M_{fact} suffices for the fixed decoder to make posterior-decoded and sampler-decoded distributions close, but it is not by itself a certificate that decoded samples match the data distribution. Conversely, a large value does not force a visible failure, since a decoder can be insensitive to the mismatched directions and map distinct latent distributions to nearly the same images. Whether a trained decoder actually conditions on the leaked structure is an empirical question that the decomposition leaves open, which is why the intervention in Section is needed rather than redundant.

Diagnostics

Each term gets a finite-sample proxy. We measure the proxies, not the information quantities, and never estimate M_{fact} as a whole. **Term 1** is proxied by global MMD, $\text{MMD}^2(q(z_s), \mathcal{N}(0, I))$. **Term 2** by inter-class style separation $\Delta_{\text{inter}} = \binom{K}{2}^{-1} \sum_{j < k} \|\bar{\mu}_s^{(j)} - \bar{\mu}_s^{(k)}\|_2$ and by linear-probe accuracy $\text{LP}(z_s \rightarrow y)$. Δ_{inter} is a mean-separation statistic and thus a convenient symptom, not what Proposition 1 is about: that proposition concerns dependence in general and holds even when class means coincide. An HSIC-based proxy would also apply here, but our fixed-bandwidth estimator saturates at z_s 's empirical scale, returning a near-constant value across checkpoints whose Δ_{inter} and LP vary widely, so we do not report it. **Term 3** is already computed each run as $\mathcal{L}_{\text{class}}$. **Term 4** has no existing proxy; we compare real within-class pairs against independently recombined ones,

$$\text{JointMMD} = \frac{1}{K} \sum_k \text{MMD}^2(\{(z_c^i, z_s^i)\}_{y_i=k}, \{(z_c^i, z_s^{\pi(i)})\}_{y_i=k}), \quad (2)$$

where π permutes style codes within class k , breaking the within-class coupling while leaving both per-class marginals, and hence terms (1)–(3), untouched.

Method and Case Study

The standard style regularizer, Eq. 1, matches only the marginal. The remedy we study instead matches each class conditional:

$$\mathcal{L}_{\text{style-cls}} = \frac{1}{K} \sum_{k=1}^K \text{MMD}_{\text{RBF}}(\{z_c^i : y_i=k\}, \{\varepsilon_j\}), \quad (3)$$

$\varepsilon_j \sim \mathcal{N}(0, I)$. Eq. 1 has no gradient incentive to reduce inter-class separation, since the mixture can match $\mathcal{N}(0, I)$ while every conditional stays class-structured; Eq. 3 penalizes exactly that. Nothing about it is specific to our architecture: it needs a labeled batch and a style latent. In F-CS-WAE, the semantic MMD and auxiliary classifier already consume the same labels, so adding Eq. 3 does not change the supervision regime or annotation budget. Applied during training to an otherwise unsupervised model, it would require labels or pseudo-labels; this is distinct from the diagnostics of Section , which need labels only on the evaluation set and apply post hoc to any trained model.

F-CS-WAE instantiates the setup concretely: a hyper-spherical semantic variable $z_c \in \mathbb{S}^{d_c-1}$ aligned to class-conditional Spherical Cauchy priors, and a Euclidean style variable $z_s \in \mathbb{R}^{d_s}$ regularized by Eqs. 1–3, with a shared ResNet-18 trunk feeding two heads and a residual upsampling decoder ($d_c=64, d_s=128$). The objective adds a supervised semantic MMD, an aggregated semantic MMD, and an auxiliary classifier on μ_c that prevents semantic collapse; these shape z_c and are not part of the leakage remedy. Coefficients ramp over a four-phase 300-epoch curriculum in which per-class style regularization begins in phase B, after a 50-epoch reconstruction-only phase A during which the decoder does see real (z_c, z_s) pairs unregularized. Section tests whether removing that warmup matters; it does not. Full details are in the supplementary material.

Table 1: CIFAR-10 against baselines matched in backbone (ResNet-18) and latent dimension ($d=192$), grouped by how the label is used. F-CS-WAE is 3-seed mean. The fair clustering comparison is against the label-guided group, which like F-CS-WAE pushes the label into the latent.

Method	ACC	NMI	FID
F-CS-WAE	0.809	0.650	83.0
<i>Label-guided (label shapes the latent)</i>			
AEWithCE	0.665	0.611	144.3
AEWithSupCon	0.541	0.599	275.5
AEWithTriplet	0.506	0.502	281.4
AEWithCenterLoss	0.243	0.122	235.8
<i>Conditional-generative (label $\rightarrow$ decoder)</i>			
ConditionalVAE	0.133	0.024	396.4
ConditionalWAE-MMD	0.102	0.008	170.3
GaussianClassPriorWAE	0.103	0.008	174.7
<i>Unsupervised</i>			
ResNetAE	0.220	0.102	121.0

At generation we either sample naively, $z_c \sim p(z_c | y=k)$ and $z_s \sim \mathcal{N}(0, I)$, or apply a post-hoc class-conditional style prior: estimate $\bar{\mu}_s^{(k)}$ and $\bar{\sigma}_s^{2(k)}$ from encoded training data and draw $z_s \sim \mathcal{N}(\bar{\mu}_s^{(k)}, \tau^2 \text{diag}(\bar{\sigma}_s^{2(k)}))$ with $\tau=0.25$. The second needs no retraining, only a forward pass.

Experiments

CIFAR-10 (Krizhevsky 2009) is the primary clustering/generation benchmark; MNIST (LeCun et al. 1998) and Fashion-MNIST (Xiao, Rasul, and Vollgraf 2017) are used for diagnostics. We report clustering (ACC/NMI/ARI), reconstruction (SSIM/LPIPS (Zhang et al. 2018)), sample quality (FID (Heusel et al. 2017), 10k vs 10k), and generated-class accuracy (Gen-ACC), the fraction of generated images assigned to the sampled class by an independently trained external classifier.

Before using F-CS-WAE to study leakage we check it is not degenerate, since a degenerate model could leak trivially. Table 1 compares it against eight baselines matched in backbone and latent dimension: seven use labels and ResNetAE is unsupervised. Against the label-guided autoencoders, the fair comparison, F-CS-WAE (80.9%) exceeds the best (66.5%) and has the lowest FID in the table. The conditional-generative baselines cluster near chance because their label enters only at the decoder, leaving the encoder latent unshaped; they are included as the closest generative analogues, not as a clustering comparison. This is evidence of competitiveness, not a controlled superiority claim: the baselines are single-seed, F-CS-WAE uses both latent-shaping and a generative prior where each baseline uses one, and we did not equalize per-method tuning.

Evaluation independence. All generated-class metrics in the main paper are scored directly from generated pixels by a classifier trained only on real images and independent of the model under test: a 4-layer CNN for MNIST (99.4% real-test accuracy) and a WideResNet-28-10 (Zagoruyko and Komodakis 2016) for CIFAR-10 (94.1%). Under this fixed pro-

Table 2: Conditional style leakage across a case-study model and four reference latent baselines on MNIST. Global MMD is near zero for five of the six rows; WAE-MMD is the exception. Δ_{inter} and LP remain large. Gen-ACC is scored by the external CNN and uses naive $z_s \sim \mathcal{N}(0, I)$. The four reference families share one architecture and budget; F-CS-WAE uses its native configuration, so conclusions are within-row rather than ranked across rows.

Model	MMD	Δ_{inter}	LP	Gen-ACC
VAE	0.0002	2.86	87.4%	N/A
WAE-MMD	0.0576	9.38	93.0%	N/A
β -TCVAE	0.0002	2.30	73.9%	N/A
FactorVAE	0.0009	2.27	75.9%	N/A
F-CS-WAE	0.0013	6.27	100.0%	0.16
F-CS-WAE + per-class	0.0015	1.22	42.6%	0.66

ocol, MNIST naive Gen-ACC is 0.16, the class-conditional prior reaches 0.97, and latent-swap style-following is 96.5%. On CIFAR-10, the class-conditional diagonal prior at $\tau=0.25$ reaches 0.41 and the empirical style bank reaches 0.88. Internal-classifier scores are reported only in the supplementary material as a robustness comparison; they are not mixed with the primary results below.

Marginal metrics hide the leakage

Table 2 applies the diagnostic to four unsupervised baselines and F-CS-WAE on MNIST. The baselines are purpose-built for this comparison and share the same encoder family, latent dimension, optimizer, and 100-epoch budget with one another. F-CS-WAE is evaluated in its native configuration. Accordingly, the table is an audit rather than a leaderboard: for every row individually, global MMD is decoupled from Δ_{inter} and LP, and the argument does not depend on ranking leakage severity across model families.

All four baselines show LP far above chance. Three do so with near-zero global MMD; WAE-MMD is the exception, with the largest global MMD (0.0576, some 30–300 $\times$ the others), suggesting its own regularizer had not converged, and correspondingly the largest baseline Δ_{inter} and LP. More telling are β -TCVAE and FactorVAE: their objectives explicitly penalize latent dependence, and they do achieve the smallest Δ_{inter} of the five, yet LP stays at 74–76%. Penalizing total correlation among latent *dimensions* reduces, but does not remove, dependence between the latent and the *label*. These are different quantities.

F-CS-WAE without the remedy is the most extreme case ($\Delta_{\text{inter}} = 6.27$, LP= 100%), and the consequence is concrete rather than statistical. On the same checkpoint it reaches 99.15% clustering accuracy and SSIM 0.983, yet class-conditional generation under naive sampling reaches only 16% external Gen-ACC. The failure is also structured rather than random: probability mass collapses onto a handful of attractor digits regardless of the class requested, which is what one expects if the decoder resolves an unfamiliar (z_c, z_s) pair by trusting whichever code it learned to read identity from. A 2D t-SNE of z_s makes the leakage visually obvious only for

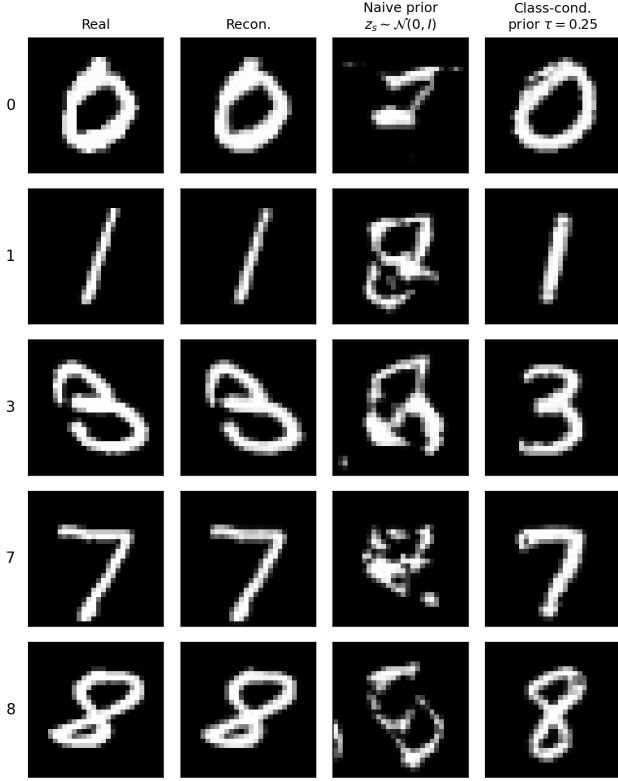


Figure 1: Real image, posterior reconstruction, naive-prior sample, and class-conditional-prior sample, for five digits. Reconstructions are indistinguishable from the real image (SSIM 0.983); naive-prior samples are legible digits but frequently the *wrong* one; the class-conditional prior restores the intended identity.

the two largest- Δ_{inter} rows; for VAE, β -TCVAE, and FactorVAE it is detected by LP but invisible in projection, which is a reason to prefer the quantitative diagnostic over a plot.

The decoder reads identity off the style code

A probe shows z_s contains class information; it does not show the decoder *uses* it. For every ordered class pair (a, b) we take μ_c from a real test image of class a and μ_s from an independent real image of class b , decode, and classify. This uses only real, individually-encoded outputs rather than prior draws, so a failure to recover class a cannot be blamed on an out-of-support prior sample, though the pairing itself may lie off the joint support the decoder trained on.

At $\delta=0$ on MNIST, external style-following is 96.5% (Figure 2): the decoder reads identity predominantly from z_s . Every row of the swap grid (fixed z_c) looks nearly identical; the column, the z_s donor, determines the digit. The per-class remedy shifts the qualitative dependence toward content on both datasets. Exact rates under the model’s internal classifier are provided only in the supplementary robustness comparison and are not used as primary evidence here.

Per-class conditional MMD confirms the same structure behind Δ_{inter} ’s single scalar: at $\delta=0$, mean per-class

Latent swap: row = z_c class (content), col = z_s class (style) — mnist_delta0

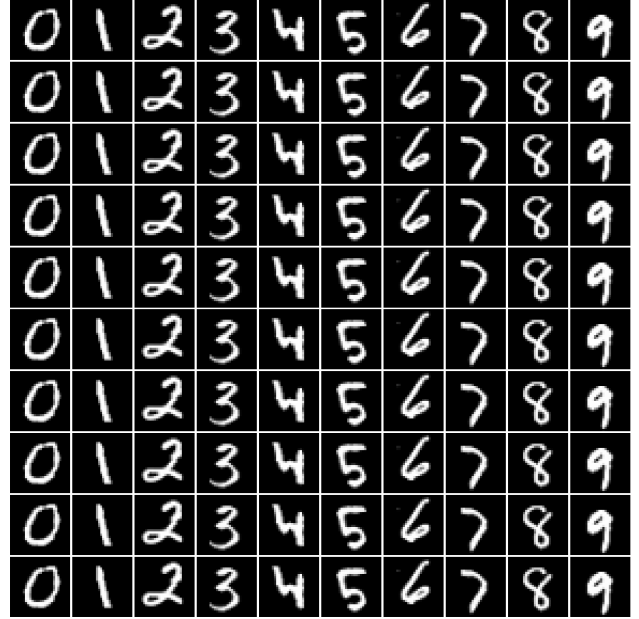


Figure 2: MNIST latent swap at $\delta=0$: rows donate z_c (content) and columns donate z_s (style). The external CNN assigns the output to the style-donor class in 96.5% of off-diagonal swaps. Internal-classifier heatmaps for both δ settings are retained only as a supplementary robustness comparison.

$\text{MMD}^2(q(z_s | y=k), \mathcal{N}(0, I))$ is $21\times$ the global MMD on MNIST (0.0186 vs 0.00089) and $15\times$ on CIFAR-10. Every one of the ten per-class values sits far above the global figure the marginal regularizer actually optimizes. At $\delta=1$ the gap narrows but does not close ($4.4\times$; $5.3\times$).

Is this an off-support artifact? The obvious objection is that cross-class pairs simply leave the training joint’s support. Grading the intervention by support answers it. Same-class cross-image pairs sit at $1.2\times$ the same-image kNN distance to the training joint and preserve identity, whereas cross-class pairs sit at $2.1\times$ and largely lose the content-donor identity. Interpolating z_s between a same-class and a cross-class donor hands identity over monotonically, crossing at $\alpha \approx 0.38$, well before the pair reaches cross-class support distance. Style-dominance turns on as soon as z_s carries a competing class signal, not only once the pair leaves the support. Cross-class pairs are measurably farther out, so the objection is not fully dissolved, but the effect does not reduce to that distance.

What the remedies do and do not fix

Per-class style MMD works in the intended direction without finishing the job. It cuts Δ_{inter} by 81% ($6.27 \rightarrow 1.22$) and LP by 57.4 points ($100\% \rightarrow 42.6\%$), still far above the 10% floor and higher than three of the four unsupervised baselines.

Table 3: Representation-level remedies on MNIST at matched backbone, dimensions, and training budget. Generation scores from the model’s internal classifier are moved to the supplementary robustness comparison. Single-seed point estimates are reported.

Remedy	LP	Δ_{inter}	JointMMD
None ($\delta=0$)	100.0	6.27	.0037
Per-cls. MMD ($\delta=1$)	42.6	1.22	.0043
VFAE-style cond.	46	1.25	.0042
HSIC penalty	37	1.08	.0041
Gradient reversal	21	0.82	.0044
+ joint term-4	40	1.15	.0016

Table 4: Six-point sweep of the per-class style MMD weight δ . Single-seed point estimates are reported. Δ_{inter} decreases steadily on both datasets, while ACC is non-monotonic and has one pronounced dip per dataset.

δ	MNIST		CIFAR-10	
	ACC	Δ_{inter}	ACC	Δ_{inter}
0	.992	6.21	.813	3.74
0.03	.994	3.27	.811	2.48
0.1	.990	2.12	.810	2.03
0.3	.994	1.55	.715	1.59
1	.859	1.22	.800	1.05
3	.993	1.18	.799	0.95

Theorem 1 explains why this term-2 remedy is necessarily partial: Eq. 3 drives terms (1)–(2) down by construction but places no pressure on term (4), the within-class dependence a decoder can still exploit once every $q(z_s | y=k)$ individually matches the prior.

Table 4 sweeps six values. Δ_{inter} decreases steadily on both datasets, so the leakage-reduction mechanism behaves as expected across the range rather than only at the endpoints. Naive FID broadly improves but is not monotonic, and clustering ACC has a pronounced single-seed dip at $\delta=1$ on MNIST and $\delta=0.3$ on CIFAR-10. With one seed per point, we cannot attribute those dips to δ rather than initialization and make no significance claim across sweep settings.

Because per-class MMD only reduces term 2 to 42.6% LP, it cannot separate “term 2 is still too large” from “term 2 is not the whole story”. Table 3 therefore compares four remedies that attack term 2 through different mechanisms, plus one that also targets term 4. Gradient reversal is much the strongest term-2 remedy, driving LP to 21%, within 11 points of chance and less than half what per-class MMD achieves. Yet every term-2-only intervention leaves JointMMD in the narrow 0.0041–0.0044 range. Targeting term 4 directly cuts JointMMD by 2.7 $\times$, the only intervention that moves this proxy substantially, while its LP remains higher than gradient reversal’s. Thus reducing $I(z_s; y)$ does not by itself reduce within-class $z_c - z_s$ dependence, exactly the distinction made by Corollary 1. Internal-evaluator generation scores for these checkpoints are retained only as a robustness comparison in

Table 5: Single-variable perturbations from the $\delta=0$ checkpoint, each removing one candidate confound, both datasets. Single-seed point estimates are reported.

Perturbation	MNIST		CIFAR-10	
	Δ_{inter}	LP	Δ_{inter}	LP
None (baseline)	6.27	100.0	3.80	86.9
Balanced dims	4.79	99.9	2.11	52.4
Smaller style	3.28	92.6	1.68	44.0
No warmup	5.53	100.0	3.78	88.3
Gaussian prior	5.35	99.6	3.82	74.5
vMF prior	5.69	100.0	3.03	83.0
No classifier	5.09	97.1	3.15	67.1

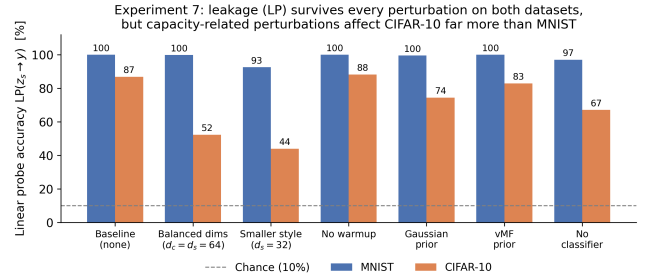


Figure 3: Linear-probe recovery of the label from z_s for the baseline and six perturbations. Leakage survives all of them on both datasets, but the two capacity perturbations move CIFAR-10 far more than they move MNIST, while no-warmup and the vMF prior barely move either.

the supplementary material.

We calibrate the term-4 proxy against a within-class double-permutation null using 200 permutations. Every checkpoint sits 4–5 $\times$ above its null ($p=0.005$), so term 4 is genuinely non-zero, while the $\delta=0 \rightarrow 1$ change is not significant ($p=0.31$): per-class style MMD leaves term 4 statistically untouched even as it cuts Δ_{inter} and LP sharply. A conditional-HSIC cross-check agrees. One comparison we avoid: MNIST and CIFAR-10 JointMMD values are close, but even after standardization, equal MMD on two different latent distributions does not imply equal dependence, so we draw no cross-dataset conclusion from it.

A five-seed replication confirms the remedy pattern on both datasets (Supplementary Section J.1). On CIFAR-10, gradient reversal gives LP $27 \pm 7\%$ with JointMMD $.0039 \pm .0006$, whereas the joint remedy gives LP $34 \pm 7\%$ but lowers JointMMD to $.0016 \pm .0004$. The corresponding MNIST values are $24 \pm 5\%$ /.0043 $\pm$.0005 and $41 \pm 6\%$ /.0017 $\pm$.0003, respectively.

Transfer and robustness

Several design choices in F-CS-WAE could plausibly be the real explanation for why z_s absorbs so much class information: the style latent being larger than the semantic one, the reconstruction-only warmup, the prior geometry, or the auxiliary classifier. Table 5 and Figure 3 perturb each independently on both datasets. LP never drops below 92.6%

Table 6: Generation under z_s sampling strategies on the $\delta=0$ checkpoints (z_c always from the class prior). Gen-ACC uses the fixed external CNN on MNIST and WideResNet-28-10 on CIFAR-10.

Style sampling z_s	MNIST		CIFAR-10	
	Gen-ACC	Div.	Gen-ACC	Div.
Global $\mathcal{N}(0, I)$	0.16	0.151	0.09	0.187
Class diag., $\tau=0.25$	0.97	0.135	0.41	0.101
Empirical style bank	0.96	0.182	0.88	0.241

on MNIST or 44.0% on CIFAR-10, and neither approaches chance. On MNIST every perturbation is mild, costing at most 7.4 points; on CIFAR-10 the same six span a 44-point range. Smaller style capacity is the largest effect on both (7.4 and 42.9 points), followed by balanced dims, but capacity is not uniquely special: removing the classifier costs 19.8 points on CIFAR-10 and the Gaussian prior 12.4, while no-warmup and the vMF prior barely move it.

Two readings follow. Capacity is a real, dataset-dependent contributor, consistent with severity tracking how much class structure a dataset forces into z_s , and a paper that checked only MNIST would have badly understated how much one hyperparameter can move this number. We also flag a confound we do not disentangle: the perturbations that reduce LP most also reduce representational capacity, and removing the classifier collapses clustering from 99% to 55.4%, so part of each drop may reflect a weaker representation rather than genuine invariance. Even so, no single tested perturbation drives LP near chance, so none of these choices *eliminates* the leakage, even though capacity clearly modulates it.

Leakage severity also follows the expected dataset ordering. Measuring intra-class diversity independently of our model, as mean within-class pairwise cosine distance in a frozen DINOv2 space, confirms the ordering with clear separation: MNIST 0.33, Fashion-MNIST 0.47, CIFAR-10 0.66. Both Δ_{inter} and LP decrease along it (MNIST 6.27/100%, Fashion-MNIST 4.86/97.7%, CIFAR-10 3.80/86.9%). We do not use Gen-ACC as a cross-dataset proxy for leakage severity: unlike Δ_{inter} and LP, it is measured after decoding and therefore also inherits dataset difficulty, sample quality, and external-classifier error.

The repair does not transfer. On MNIST, the external CNN raises Gen-ACC from 0.16 under global Gaussian sampling to 0.97 under the class-conditional diagonal prior at $\tau=0.25$ (Table 6). On CIFAR-10, the same parametric recipe reaches only 0.41 under the external WideResNet. The empirical style bank reaches 0.88 and also has higher measured diversity (0.241 versus 0.101), indicating that matching only a unimodal diagonal Gaussian misses important shape in $q(z_s | y=k)$. Controls for mean shift, variance shrinkage, and shuffled labels show the same qualitative mechanism under the internal evaluator and are reported only as a supplementary robustness comparison. The diagnostic therefore transfers across datasets more cleanly than the parametric repair does.

The five-seed replication yields external Gen-ACC $0.16 \pm$

0.03 , 0.96 ± 0.02 , and 0.95 ± 0.02 on MNIST, and 0.10 ± 0.03 , 0.43 ± 0.07 , and 0.86 ± 0.05 on CIFAR-10 for global, class-conditional diagonal, and empirical-bank sampling, respectively (Supplementary Section J.2). Thus the negative transfer is not specific to the single checkpoints in Table 6.

Discussion

CIFAR-10’s higher intra-class diversity weakens conditional style dependence, and Δ_{inter} /LP confirm this cleanly, but the practical remedy does not inherit that cleanliness. CIFAR-10 samples stay visibly blurry (FID ~ 83 –84), and we avoid calling naive output “visually plausible”; at this FID the accurate description is recognizable but not sharp. Global aggregate matching is standard in WAEs and disentanglement models, and our results show it is insufficient whenever the style variable must be class-invariant for generation. Per-class style MMD is best read as a supervised analogue of total-correlation penalties, aimed at the single dependency $z_s \not\perp y$ rather than at dependence among latent dimensions in general. Finally, the MNIST checkpoint reaches SSIM 0.983 yet FID 73.99 under naive sampling: reconstruction measures fidelity on the posterior manifold, not the prior, and evaluation of generative clustering should report both, since the two can diverge substantially.

Limitations. We report no component-wise ablation beyond the δ axis. The per-class remedy assumes labels already available to the label-guided training objective and is not, by itself, an unsupervised disentanglement loss. Gen-ACC is confounded by dataset-intrinsic difficulty. Although five-seed replication confirms the CIFAR-10 transfer failure and the remedy ordering, we have not tested whether a richer conditional (a per-class mixture) closes the gap. The robustness check perturbs one variable at a time, so joint perturbations, which could compound given how much larger each effect is on CIFAR-10, remain untested. Three datasets remain too few to treat the diversity–leakage relationship as a measured correlation rather than a consistent ordering.

Conclusion

Conditional style leakage is a real failure mode in factorized generative models: aggregate regularization of a style variable does not prevent it from becoming class-dependent, because marginal matching is compatible with z_s being maximally informative about the label rather than merely correlated with it. We placed it inside an exact decomposition of what factorized sampling requires, measured it across a case-study model and four reference latent baselines, and showed that a decoder trained this way is measurably more sensitive to the leaked code than to the semantic one. No single perturbation of capacity, curriculum, prior geometry, or supervision eliminates the leakage on either dataset, though capacity strongly modulates its magnitude on CIFAR-10. Comparing four mechanistically different term-2 remedies settles what one remedy could not: they span 21–46% probe recovery yet leave the term-4 proxy unchanged, whereas the joint remedy cuts that proxy by $2.7\times$ despite a higher probe score than gradient reversal. Eliminating conditional style leakage is necessary but not sufficient, exactly as Corollary 1

states. Under the fixed external protocol, the conditional prior reaches 0.97 Gen-ACC on MNIST but only 0.41 on CIFAR-10; an empirical bank reaches 0.88. Thus fixing a diagnostic finding on one dataset does not fix the underlying generative gap on another. Factorized generative models should verify class-invariance of style variables with conditional diagnostics rather than marginal statistics, and should report where their remedies do and do not generalize.

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